Foundation species have strong effects on their communities as a result of their large size or high abundance. Examples of foundation species include trees, desert shrubs such as creosote, and certain marine algae, such as kelp. Such species often have community-wide effects because they provide habitat or food. Foundation species may also be competitively dominant—superior in exploiting key resources such as space, water, nutrients, or light.

The impact of a foundation species can be discovered when it is removed from the community. For example, the American chestnut was a competitively dominant tree in deciduous forests of eastern North America before 1910, making up more than 40% of mature trees. Then humans accidentally introduced the fungal disease chestnut blight to New York City via nursery stock imported from Asia. Between 1910 and 1950, this fungus killed almost all of the chestnut trees in eastern North America. In this case, removing the foundation species had a relatively small impact on some species but large effects on others. Oaks, hickories, beeches, and red maples that were already present in the forest increased in abundance and replaced the chestnuts. No mammals or birds seemed to have been harmed by the loss of the chestnut, but seven species of moths and butterflies that fed on the tree became extinct.

In contrast to foundation species, **keystone species** are not usually abundant in a community. They exert strong control on community structure not by numerical might but by their pivotal ecological roles. **Figure 54.20** highlights the importance of a keystone species, a sea star, in maintaining the diversity of an intertidal community. In this case, the sea star affects its community by feeding on and there by limiting the abundance of a competitively dominant species, a mussel.

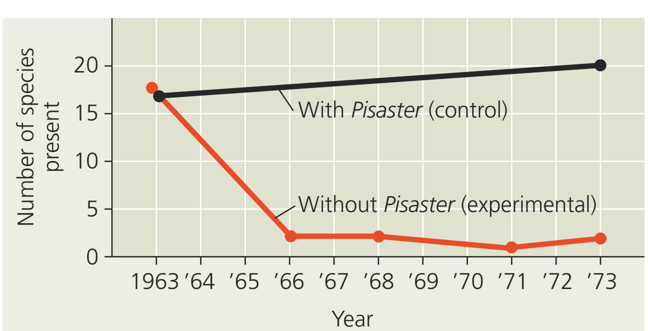
Still other organisms exert their influence on a community not through trophic interactions but by changing the physical environment. Species that create or dramatically alter their environment are called **ecosystem engineers**. A familiar ecosystem engineer is the beaver **(Figure 54.21)**. The effects of ecosystem engineers on other species can be positive or negative, depending on the needs of the other species. Some foundation species, such as trees, can also be considered ecosystem engineers because their presence modifies the physical environment in ways that create habitats on which other species depend.

▼ Figure 54.20 Inquiry

Is *Pisaster ochraceus* a keystone species?

Experiment In rocky intertidal communities of western North America, the relatively uncommon sea star *Pisaster ochraceus* preys on mussels such as *Mytilus californianus*, a dominant competitor for space. Robert Paine, of the University of Washington, removed *Pisaster* from an area in the intertidal zone and examined the effect on species richness.

Results In the absence of *Pisaster*, species richness declined as mussels monopolized the rock face and eliminated most other invertebrates and algae. In a control area where *Pisaster* was not removed, species richness changed very little.



Conclusion *Pisaster* acts as a keystone species, exerting an influence on the community that is not reflected in its abundance.

Data from R. T. Paine, Food web complexity and species diversity, *American Naturalist* 100:65–75 (1966).

WHAT IF? *Suppose that an invasive fungus killed most individuals of* Mytilus *at these sites. Predict how species richness would be affected if* Pisaster *were then removed.*

▼ Figure 54.21 Beavers as ecosystem engineers. By felling trees, building dams, and creating ponds, beavers can transform large areas of forest into flooded wetlands.

Bottom-Up and Top-Down Controls

The ways in which adjacent trophic levels affect one another can be useful for describing community organization. These effects occur in two general ways: organisms can be controlled by what they eat ("bottom-up" control) or by what eats them ("top-down" control).

In **bottom-up control**, the abundance of organisms at each trophic level is limited by nutrient supply or the